

Multiple regression



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Linear model recap

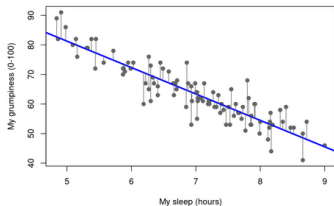
The linear model

- ▶ Assumes that a continuous variable y can be meaningfully predicted as a function of a weighted linear combination of predictors x_i
- ▶ In the special case of ANOVA the predictors are of type factor; in the special case of regression they are of type numeric
- ▶ Linear models can also combine predictors of type numeric and factor ("ANCOVA")
- ▶ Predictors of type factor are assumed to be orthogonal, whereas numeric predictors may correlate with one another, especially if they are observed/measured rather than manipulated.

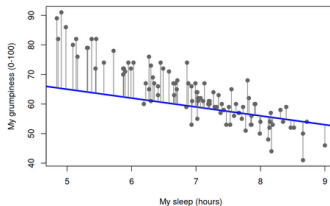
$$y_i \sim \text{Normal}(\mu_i, \sigma^2)$$

$$\mu_i = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \dots + \beta_n x_{in}$$

Linear model: Graphical



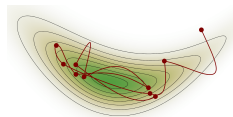
(a) Residuals for the best fitting line.



(b) Residuals for a poor line.

- ▶ $\beta_0 = \alpha$ = "intercept": point at which the regression line crosses the y-Axis when all $x_i = 0$.
- ▶ β_1 = slope or inclination of the line.
- ▶ Goal: Find values for β_0 and β_1 that minimise the sum of the squared residuals, $\sum (y_i - \hat{y}_i)^2$.

Minimising squared differences



- ▶ Simple frequentist regression: Exact computation via "ordinary least squares" (OLS)
- ▶ Complex (e.g. multilevel) frequentist regression: Estimation via Maximum Likelihood (ML)
- ▶ Simple Bayesian regression: Exact computation via Bayes theorem & Bayesian conditionalization
- ▶ Complex Bayesian regression: Approximation via Markov-Chain-Monte-Carlo (MCMC)

Formal notation

Mathematical model notation

Two alternative notations

Distributions for all parameters

- ▶ $y_i \sim \text{Normal}(\mu_i, \sigma^2)$
- ▶ $\mu_i = \beta_0 + x_{i1}\beta_1 + x_{i2}\beta_2 + x_{i1}x_{i2}\beta_3 + \dots$
- ▶ **Priors** for each β and σ

Distribution for residuals

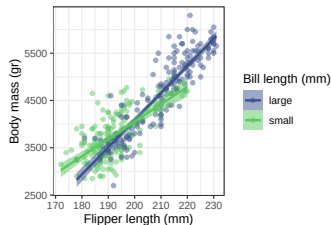
- ▶ $y_i = \beta_0 + x_{i1}\beta_1 + x_{i2}\beta_2 + x_{i1}x_{i2}\beta_3 + \dots + \epsilon_i$
- ▶ $\epsilon_i \sim \text{Normal}(0, \sigma^2)$

Formula notation in R

- ▶ `m1 <- brm(y ~ 1, data = dat)`
(Intercept-only model)
- ▶ `m1 <- brm(y ~ x1, data = dat)`
(model with main effect of `x1`)
- ▶ `m1 <- brm(y ~ x1 + x2, data = dat)`
(model with two main effects)
- ▶ `m1 <- brm(y ~ x1 + x2 + x1:x2, data = dat)`
`m2 <- brm(y ~ x1*x2, data = dat)`
(model with two main effects & their interaction, `m1 = m2`)

Example with brms

Penguin example



To what extent can be a penguin's body mass be predicted from the length of its bill and of its flippers?

```
1 > library(brms)
2 > m1 <- brm(body_mass ~ bill_len + flipper_len
3             , data = penguins
4             )
```

Model notation with priors

- ▶ The linear model (c.v. the likelihood in Bayes theorem)
 - $y_{i_{body}} \sim \text{Normal}(\mu_{i_{body}}, \sigma^2)$
 - $\mu_{i_{body}} = \beta_0 + x_{i_{flipper}}\beta_1 + x_{i_{bill}}\beta_2$
- ▶ Priors for the parameters:
 - $\beta_0 \sim \text{Normal}(0, 10)$
 - $\beta_1 \sim \text{Normal}(0, 5)$
 - $\beta_2 \sim \text{Normal}(0, 5)$
 - $\sigma^2 \sim \text{Normal}_+(0, 3)$

Default priors with brms

What are the default priors, and what types of priors are there at all, given our model formula?

```
1 > library(brms)
2 > default_prior(body_mass ~ bill_len + flipper_len
3 , data = penguins
4 )
5           prior      class      coef group resp dpar nlpar lb ub
6           (flat)         b              bill_len
7           (flat)         b      flipper_len
8           (flat)         b
9 student_t(3, 4050, 889.6) Intercept
10 student_t(3, 0, 889.6)      sigma              0
```

- ▶ Why may the ("flat") priors be suboptimal?
- ▶ Where does the intercept of 4050 for the intercept come from, and why might it be suboptimal?

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```

- ▶ Why may the ("flat") priors be suboptimal?
- ▶ Where does the intercept of 4050 for the intercept come from, and why might it be suboptimal?
- ▶ By default **brms** uses the data (e.g. variable means) to set its default priors → double-dipping problem.

Own priors with brms

Which priors make sense taking into account our knowledge of the subject matter, but without restricting the space of possible outcomes too much?

► Example of own prior for `bill_len`:

```
1 > myPriorForB <- c(prior(normal(0, 5), class = b, coef = bill_len))
2
3 > m1 <- brm(body_mass ~ bill_len + flipper_len
4             , data = penguins
5             , prior = myPriorForB
6             )
```


"Proper priors"

- ▶ **Proper priors** are characterised by two features:
 - All values of probability distributions are ≥ 0 .
 - The values sum up to 1 (for discrete distributions) or integrate to 1 (for continuous distributions).
- ▶ Proper priors respect the axioms of probability theory, which allows them to be meaningfully interpreted.
- ▶ In most cases, proper priors lead to proper posteriors.
- ▶ **Goal:** To find priors that reflect plausible parameter values and plausible assumptions about the uncertainty around these parameter values.

(Veenman et al., 2023).

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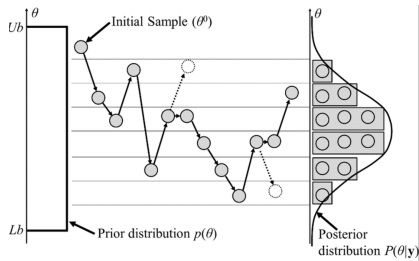
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([Veenman et al., 2023](#)).

Estimation with MCMC

MCMC



- Roughly, MCMC involves estimating the denominator of Bayes theorem (the probability of the data across all parameters) by taking repeated samples from the posterior with replacement. Values sampled more often indicate areas of the parameter space with higher posterior density.

Features of MCMC

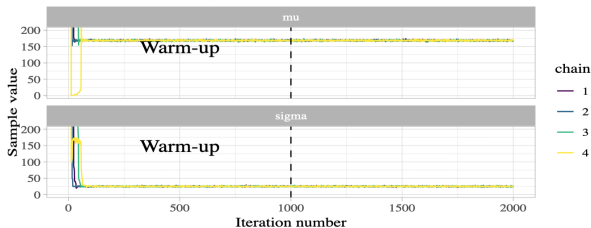
- ▶ In theory, the value of each sampled element depends only on the previous value, not on values before that in the "chain" (this is the "**Markov property**").
- ▶ However, in practice samples can still correlate with one another over longer sequences (called "**autocorrelation**"), especially if the MCMC algorithm moves around the parameter space in small steps.
- ▶ Autocorrelation is visible in the MCMC chain as a drift in one direction or another. This can bias parameter estimation and make results difficult to interpret.

Chain convergence

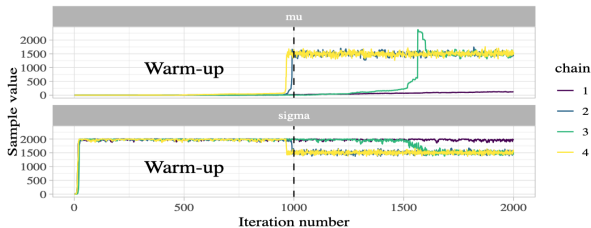
- ▶ When several MCMC chains are run in **parallel**, their values can be merged to reduce bias. Typical default: 4.
- ▶ The more samples are drawn, the more precise the resulting estimate will be. Typical default: 10000.
- ▶ Autocorrelation is a stronger issue at the beginning of the MCMC process, when the values sampled still depend on the starting values. Therefore the starting values are often removed ("**warmup**" or "**burnin**").
- ▶ Chains that converge well lie on top of one another with no drift, showing instead a horizontal "zig-zag" or "hairy caterpillar" pattern.

Convergence graphical

a. Model that converged.



b. Model that did not converge.

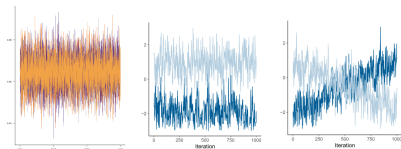


MCMC in brms

```
1 # General settings
2 options(mc.cores = parallel::detectCores())
3
4 # Formula with MCMC settings
5 m1 <- brm(body_mass ~ bill_len + flipper_len
6           , data = penguins
7           , warmup = 1000
8           , iter = 5000
9           , chains = 4
10           , cores = 4
11           )
```

Some MCMC parameters can be specified directly in brms, inside the brm function.

Convergence diagnostics



- **Trace plot:** Do they "mix well" and show a "hairy caterpillar" pattern?
- **Gelman–Rubin diagnostic $\hat{R}$:** Is the ratio of the variance between chains to the variance within chains not above 1.05?
- **Effective sample size (ESS):** Is the estimated number of fully independent samples not below 400 (or minimally 100) for any parameter?

(Veenman et al., 2023; Vehtari et al., 2021).

Convergence diagnostics with brms: numerical

Model: `m1 <- brm(body_mass ~ bill_len + flipper_len)`

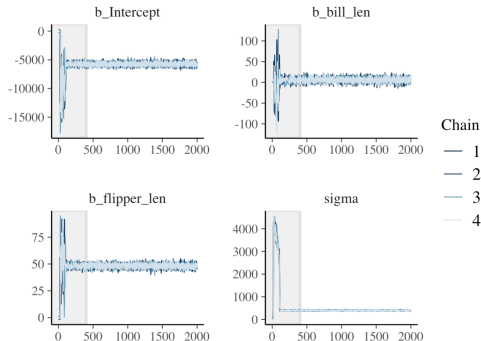
```
> m1$fit
Inference for Stan model: anon_model.
4 chains, each with iter=2000; warmup=1000; thin=1;
post-warmup draws per chain=1000, total post-warmup draws=4000.
```

	mean	se_mean	sd	2.5%	25%	50%	75%	97.5%	n_eff	Rhat
b_Intercept	-5738.80	4.73	314.21	-6364.65	-5944.75	-5735.40	-5533.23	-5119.05	4408	1
b_bill_len	6.02	0.10	5.35	-4.73	2.48	5.89	9.53	16.55	2893	1
b_flipper_len	48.16	0.04	2.06	44.15	46.78	48.11	49.49	52.26	3056	1
sigma	395.56	0.25	15.00	367.70	385.26	394.95	405.61	425.78	3733	1
Intercept	4201.47	0.36	21.75	4159.68	4186.50	4201.17	4216.89	4242.73	3686	1
lprior	-15.04	0.00	0.01	-15.06	-15.04	-15.04	-15.03	-15.02	3480	1
lp__	-2538.86	0.04	1.46	-2542.59	-2539.59	-2538.53	-2537.81	-2537.06	1532	1

Samples were drawn using NUTS(diag_e) at Sun Nov 23 22:58:41 2025.
For each parameter, n_eff is a crude measure of effective sample size,
and Rhat is the potential scale reduction factor on split chains (at
convergence, Rhat=1).

Convergence diagnostics with brms: graphical

```
1 > library(bayesplot)
2 posterior <- rstan::
  extract(m1$fit,
    inc_warmup = TRUE,
    permuted = FALSE)
3 bayesplot::mcmc_trace(
  posterior, n_
    warmup = 400, pars
    = variables(m1)[c
    (1,2,3,4)])
4 > posterior
```



Posterior parameter values

Means and credible intervals

- ▶ Provide information on the posterior predictions of the model, and on the uncertainty of these predictions.
- ▶ In numerical format: can be obtained e.g. with `summary()`, `posterior()`, and `hdi()`.
- ▶ In graphical format: can be obtained e.g. with `plot()` and `mcmc_areas()` (see `shinystan` & `bayesplot` packages).

Tables of posteriors

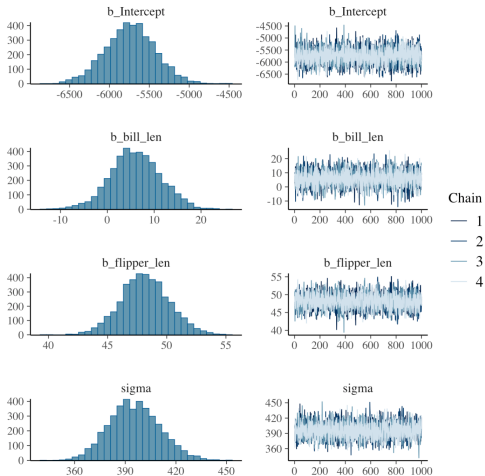
```
1 > library(kableExtra)
2 > m1_fixed <- summary(m1)[[15]][,c("Estimate", "Est.Error", "l-95% CI", "u-95% CI",
  "Bulk_ESS", "Rhat")]
3 > m1_fixed <- rownames_to_column(m1_fixed, var = "Parameter")
4 > colnames(m1_fixed) <- c("Parameter", "Mean", "SE", "Lower Bound", "Upper Bound",
  "$n_{eff}$", "$\\hat{R}$")
5 > table <- kable(m1_fixed, caption = "Fixed effects for model m1.")
6 table
```

Table 1: Fixed effects for model m1.

Parameter	Mean	SE	Lower Bound	Upper Bound	n_{eff}	$\hat{R}$
Intercept	-5739.459173	307.319017	-6340.321772	-5130.36184	17040.71	0.9999220
bill_len	5.930243	5.197813	-4.245734	16.17710	12128.09	1.0001620
flipper_len	48.183283	2.018193	44.227396	52.11286	12665.40	0.9999636

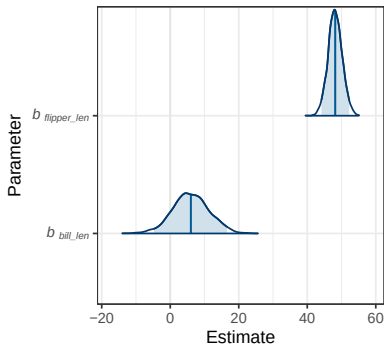
Graphs of posteriors with plot()

```
1 > plot(m1)
```



Graphs of posteriors with `mcmc_areas()` of `bayesplot`

```
1 > g2 <- mcmc_areas(m1,  
2 pars = c("b_bill_len", "b_  
3 flipper_len"),  
4 prob = 0.95, # 95% credible  
5 intervals  
6 point_est = "mean") +  
7 theme(text = element_text(  
8   size = 14)) +  
9 theme_bw() +  
10 labs(x = "Estimate", y = "  
11   Parameter") +  
12   scale_y_discrete(labels =  
13     c(expression(~italic(b[~"  
14       bill_len]")),  
15     expression(~italic(b[~"  
16       flipper_len]"))))  
17 g2
```



Model comparisons

Nested vs. non-nested models

- ▶ Two models $m1$ and $m2$ are nested when one is a special case or "subset" of the other, such that e.g. $m1$ contains all the components of $m2$, plus one or more further components.
- ▶ Models can be compared more easily when they are nested. But any models can be compared using Bayes factors.

Example (1)

Which of these models are nested?

1. $y_i \sim 1 + x_{i1}$

$$y_i \sim 0 + x_{i1}$$

2. $y_i \sim 1 + x_{i1}$

$$y_i \sim x_{i1}$$

3. $y_i \sim x_{i1} + x_{i2}$

$$y_i \sim x_{i1} + x_{i2} + x_{i3}$$

4. $y_i \sim x_{i1} + x_{i2}$

$$y_i \sim x_{i1} + x_{i3}$$

5. $y_i \sim x_{i1} + x_{i2} + x_{i1} : x_{i2}$

$$y_i \sim x_{i1} * x_{i2}$$

6. $y_i \sim x_{i1} + x_{i2} + x_{i3} + x_{i1} : x_{i3}$

$$y_i \sim x_{i1} + x_{i2} + x_{i3} + x_{i1} : x_{i2}$$

Example (1)

Which of these models are nested?

1. $y_i \sim 1 + x_{i1}$ ✓
 $y_i \sim 0 + x_{i1}$
2. $y_i \sim 1 + x_{i1}$ ✓
 $y_i \sim x_{i1}$
3. $y_i \sim x_{i1} + x_{i2}$ ✓
 $y_i \sim x_{i1} + x_{i2} + x_{i3}$
4. $y_i \sim x_{i1} + x_{i2}$
 $y_i \sim x_{i1} + x_{i3}$
5. $y_i \sim x_{i1} + x_{i2} + x_{i1} : x_{i2}$ ✓
 $y_i \sim x_{i1} * x_{i2}$
6. $y_i \sim x_{i1} + x_{i2} + x_{i3} + x_{i1} : x_{i3}$
 $y_i \sim x_{i1} + x_{i2} + x_{i3} + x_{i1} : x_{i2}$

Types of nested model comparisons

► Theory driven

- Recommended when possible, but presupposes sufficiently precise, empirically testable theories

► Data driven

- Forwards selection (from simple to complex)
- Backwards selection (from complex to simple)
- Mixed selection (combines forwards & backwards)

Types of nested model comparisons

- ▶ **Theory driven**
 - Recommended when possible, but presupposes sufficiently precise, empirically testable theories
- ▶ **Data driven**
 - **Forwards** selection (from simple to complex)
 - **Backwards** selection (from complex to simple)
 - **Mixed** selection (combines forwards & backwards)

Example (2)

Which of these sequential model comparisons uses forward selection and which uses backward selection?

1. $y_i \sim 1$
2. $y_i \sim 1 + x_{i1} + x_{i2}$
3. $y_i \sim 1 + x_{i1} * x_{i2}$

1. $y_i \sim 1 + x_{i1} * x_{i2}$
2. $y_i \sim 1 + x_{i1} + x_{i2}$
3. $y_i \sim 1$

Example (2)

Which of these sequential model comparisons uses forward selection and which uses backward selection?

Forward

1. $y_i \sim 1$
2. $y_i \sim 1 + x_{i1} + x_{i2}$
3. $y_i \sim 1 + x_{i1} * x_{i2}$

Backward

1. $y_i \sim 1 + x_{i1} * x_{i2}$
2. $y_i \sim 1 + x_{i1} + x_{i2}$
3. $y_i \sim 1$

Beispiele (3)

If we use a "book-end" selection procedure for model $y_i \sim 1 + x_{i1} + x_{i2}$, in which we compare it to a minimal and to a maximal model, what would these two comparison models look like?

Beispiele (3)

If we use a "book-end" selection procedure for model $y_i \sim 1 + x_{i1} + x_{i2}$, in which we compare it to a minimal and to a maximal model, what would these two comparison models look like?

- ▶ **Minimal:** The intercept-only model $y_i \sim 1$
- ▶ **Maximal:** The "full" model $y_i \sim 1 + x_{i1} * x_{i2}$

Model comparison with Bayes factors

$$BF_{12} = \frac{P(Y|M_1)}{P(Y|M_2)}$$

- ▶ Quantify the **relative evidence** for one model over another given the data.
- ▶ Complement the information provided by the posteriors.
- ▶ Allow nested and non-nested model comparisons.
- ▶ Naturally compensate for differences in model complexity.
- ▶ Allow us to determine that there is no evidence one way or another, so that it would be good to gather more data before drawing a conclusion.

Penguin *BF* comparison

```
1 > m1 <- brm(body_mass ~ bill_len + flipper_len
2             , data = penguins
3             , warmup = 1000
4             , iter = 5000
5             , chains = 4
6             , cores = 4
7             )
8 > m0 <- brm(body_mass ~ flipper_len
9             , data = penguins
10            , warmup = 1000
11            , iter = 5000
12            , chains = 4
13            , cores = 4
14 # Using bridge sampling
15 > bayes_factor(m0, m1)
16
17 Estimated Bayes factor in favor of m1 over m0: 20.57544
```

Although the credible interval for bill length included the Null, suggesting its impact was small, the data was around 20 times more likely under the assumption that both bill length and flipper length predict body mass (m_1), compared to the assumption that only flipper length predicts body mass (m_0).